Algorithm: Step 1 – Understand the User Prompt

Step 1: Prompt Parsing and Structured Representation

1.1 Ingest raw natural language prompt from client interface and extract core user objective.

1.2 Retrieve persistent session context and workspace history from browser IndexedDB (`PrismDB`):

- **1.2.1** Load prior interaction turns, agent memory checkpoints, and active user preferences.
- **1.2.2** Resolve local file system references, target project repositories, and access tokens.

1.3 Construct strongly-typed operational execution tuple:

$$T = \langle \text{Goal}, \text{Constraints}, \text{Context}, \mathcal{M}_{\text{tools}}, \Phi_{\text{out}} \rangle$$

- **1.3.1** Parse explicit operational constraints (deadlines, schema specifications, output formats).
- **1.3.2** Bind authorized Model Context Protocol (MCP) tool capabilities and environment permissions.

1.4 Perform input boundary and security screening:

- **1.4.1** Evaluate prompt against token limits and scan for prompt injection patterns.
- **1.4.2** Flag state-mutating actions requiring mandatory human-in-the-loop authorization.